

Bitcoin as a Liquidity Barometer Rather than a Hedge: An Empirical Re-Assessment of Popular Investment Theses

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Abstract

Since its emergence in 2009, Bitcoin has been promoted under several grand investment theses: as “digital gold” and a long-term store of value, as a hedge against inflation and monetary debasement, as a safe-haven asset negatively correlated with risk assets and sovereign bonds, and—more recently—as a financial expression of energy markets due to its proof-of-work mining mechanism. Institutional research, policy discourse, and popular commentary have all contributed to the diffusion of these narratives. [Fidelity Digital Assets+2Fidelity Digital Assets+2](#)

This paper re-examines those claims using a comprehensive panel of market data for Bitcoin (BTC), gold (GLD proxy), U.S. Treasuries, global and U.S. equity indices, and energy-sector exchange-traded funds (ETFs) from January 2012 through mid-November 2025. The analysis focuses on daily, weekly, and monthly log returns, rolling 252-day correlations, and regime-specific subsamples corresponding to key macro-financial environments: early adoption (2012–2015), pre-COVID expansion (2016–2019), the “QE-Everything” and pandemic period (2020–2021), and the tightening and post-pandemic cycle (2022 onward). It also incorporates U.S. macroeconomic variables—CPI, PPI, the effective federal funds rate, and 10-year Treasury yields—to assess whether Bitcoin behaves as an inflation hedge or macroeconomic hedge.

Across the full sample, Bitcoin exhibits only weak positive correlations with gold, U.S. Treasuries, inflation metrics, and energy-sector assets. Correlations with broad equity indices (SPY, VXUS) and global sector ETFs are consistently stronger and have risen materially in the post-2020 era. Rolling-window results show that Bitcoin’s correlation structure is highly time-varying, with sustained periods in which BTC behaves as a high-beta, pro-cyclical asset strongly positively correlated with global equities, especially during liquidity expansions and risk-on episodes.

The empirical evidence thus rejects three common theses: (1) that Bitcoin is a reliable hedge or safe haven against U.S. Treasuries, (2) that Bitcoin currently functions as a store of value analogous to gold, and (3) that Bitcoin’s price primarily reflects energy-market dynamics or electricity prices. Instead, the data support an alternative interpretation: Bitcoin behaves as a global liquidity and speculative-risk barometer whose price is highly sensitive to the availability and cost of leverage, institutional and retail risk appetite, and broader conditions in credit and equity markets. Recent events in 2024–2025—such as record inflows into spot Bitcoin ETFs, new all-time price highs above \$120,000 followed by sharp drawdowns, and growing strains in private-credit markets including the bankruptcy of First Brands Group—fit naturally within this framework. [The Esk+4Reuters+4etf.com+4](#)

1. Introduction

The emergence of Bitcoin has coincided with a period of unprecedented monetary accommodation, financialization, and technological change. Against this backdrop, investors and commentators have advanced several stylized narratives to justify allocating capital to BTC. Three have become especially prominent:

1. **Bitcoin as an anti-Treasury hedge or safe haven**, expected to appreciate when U.S. sovereign bonds and conventional financial assets are under stress.
2. **Bitcoin as “digital gold” or a long-term store of value**, mirroring or eventually replacing gold’s role in portfolios as a hedge against monetary debasement.
3. **Bitcoin as an energy-linked asset**, whose price purportedly reflects electricity costs, energy-sector dynamics, or broader commodity cycles due to its proof-of-work mining mechanism and substantial electricity demand. [CCAF+3Fidelity Digital Assets+3Fidelity Digital Assets+3](#)

These theses are not purely academic; they have tangible consequences for portfolio construction, institutional adoption, and public policy debates. Major asset managers and research houses have published reports describing Bitcoin as an aspirational store of value and potential strategic allocation in multi-asset portfolios. [Fidelity Digital Assets+2Fidelity Digital Assets+2](#) Policy authorities, including central banks and regulators, are actively assessing Bitcoin’s implications for financial stability, climate goals, and monetary sovereignty. [Nature+3CCAF+3U.S. Energy Information Administration+3](#)

Recent developments have further raised the stakes. In 2025 the United States formally established a Strategic Bitcoin Reserve and a U.S. Digital Asset Stockpile, framing confiscated Bitcoin holdings as a national reserve asset comparable to gold. [The Washington Post+5White House+5Latham & Watkins+5](#) Spot Bitcoin ETFs have attracted record inflows, propelling prices to new all-time highs above \$120,000 before a pronounced correction. [The Block+6Reuters+6etf.com+6](#) At the same time, stresses in private-credit markets—illustrated by the bankruptcy of First Brands Group and associated losses across funds at UBS, Jefferies, and other institutions—have highlighted latent fragilities in leveraged finance. [Managed Funds Association+9Private Debt Investor+9Jefferies.com+9](#)

Against this complex backdrop, it is legitimate to ask whether the early investment theses about Bitcoin still hold when confronted with a long history of realized price behavior. Rather than rely on stylized narratives, this paper grounds the discussion in data spanning over thirteen years of daily observations for Bitcoin and a representative set of other assets and macro variables.

The central findings are straightforward:

- Over the full sample (2012–2025), **Bitcoin’s daily and weekly returns exhibit only weak correlation with gold, U.S. Treasuries, and inflation metrics**, and its correlation with energy-sector equity ETFs is modest.
- **Correlations with broad equity benchmarks (SPY, VXUS) and global energy-sector indices (IXC, IYE, VDE) are consistently stronger, and have increased materially since 2020.**

- **Rolling 252-day correlations** show that BTC behaves like a high-beta, pro-cyclical risk asset during liquidity booms (2020–2021, 2023–2025), with correlations to global equities often exceeding 0.3–0.5 on a monthly basis, while relationships with gold and Treasuries remain weak or unstable.
- Monthly BTC returns show **negligible correlation with changes in CPI and PPI** and only mild relationships with interest-rate variables, suggesting that BTC is not behaving as a classical inflation hedge in the manner of inflation-linked bonds or, in some circumstances, gold. [Munich Personal RePEc Archive+4ScienceDirect+4PMC+4](#)

Taken together, these results argue against the view of Bitcoin as a reliable hedge or store of value relative to traditional safe-haven assets. Instead, Bitcoin appears better characterized as a **barometer of global liquidity and speculative risk appetite**—a volatile instrument whose price movements are amplified expressions of broader market conditions rather than diversifying hedges against them.

2. Literature Review: The Rise of Three Investment Theses

2.1 Bitcoin as a Safe Haven and Anti-Treasury Hedge

The notion of Bitcoin as a safe haven emerged early in its history, particularly among retail investors skeptical of central banks and fiat currencies. This view was reinforced by episodes where BTC rallied during perceived political or monetary turbulence and by anecdotes about capital flight from countries with capital controls or currency crises.

Academic evidence, however, is more nuanced. A stream of studies has examined whether Bitcoin functions as a hedge or safe haven relative to stocks, bonds, foreign exchange, and commodities. Bouri et al. and Shahzad et al. found that Bitcoin can exhibit *local* safe-haven behavior during short periods of extreme financial stress, particularly in tail quantiles, but not as a consistent hedge across regimes.

[ScienceDirect+1](#)

Choi and Shin, in a widely cited paper, analyzed the joint dynamics of inflation, uncertainty, and Bitcoin and gold prices using a VAR framework. They concluded that Bitcoin appreciates in response to inflation shocks, exhibiting some inflation-hedging properties, but unlike gold it does not serve as a robust safe haven during periods of heightened uncertainty. [ScienceDirect+2PMC+2](#) Rashid et al. similarly argue that Bitcoin fails to qualify as a reliable safe haven for traditional assets once volatility and structural breaks are taken into account. [Taylor & Francis Online](#)

More recent work using dynamic correlation models during the COVID-19 period finds that Bitcoin's relationship with equities and gold varies substantially over time, alternating between weak safe-haven characteristics and high-beta co-movement depending on pandemic phases and monetary conditions. [ResearchGate+2PMC+2](#) Taken together, this literature suggests that the "Bitcoin as a Treasury hedge" thesis may have been overstated or dependent on narrow event windows.

2.2 Bitcoin as "Digital Gold" and Store of Value

A second major thesis is that Bitcoin is "digital gold"—a scarce, non-sovereign asset designed to preserve value over long horizons. This narrative highlights Bitcoin's fixed supply cap of 21 million coins, halving schedule, censorship resistance, and portability. Institutional research has played a major role in promoting this view; for example, Fidelity Digital Assets' *Bitcoin Investment Thesis* and later *Bitcoin First* series describe Bitcoin as an "aspirational store of value" and "monetary good" distinct from other digital assets. [Lopp+5Fidelity Digital Assets+5Fidelity Digital Assets+5](#)

Media coverage and industry marketing have amplified these messages, frequently juxtaposing charts of gold and Bitcoin prices, using phrases such as "digital gold" and "store of value," and highlighting cross-sectional correlations during select macro episodes. Contemporary commentary often argues that Bitcoin could ultimately displace or complement gold in central bank reserves or sovereign wealth portfolios—a view now echoed in discussions around the U.S. Strategic Bitcoin Reserve and similar initiatives in other jurisdictions. [Morningstar+3Wikipedia+3The Washington Post+3](#)

Academic results again paint a more fractured picture. Several studies find that Bitcoin offers some inflation-hedging properties in specific contexts but emphasize that such benefits are highly regime- and horizon-dependent. [ScienceDirect+2ScienceDirect+2](#) Others stress Bitcoin's extreme volatility and

drawdowns, which are difficult to reconcile with the classical notion of a stable store of value. [Taylor & Francis Online+2IDEAS/RePEc+2](#)

2.3 Bitcoin as an Energy-Linked Asset

A third narrative—especially prominent among critics—is that Bitcoin is effectively an “energy derivative,” whose value is grounded in the electricity consumed by miners. Observers point to the large share of global electricity attributed to Bitcoin mining, estimated between roughly 0.2% and 0.9% of global demand in 2023, and to localized impacts on U.S. grids where cryptocurrency mining may account for 0.6–2.3% of total consumption. [Stout+8CCAF+8U.S. Energy Information Administration+8](#)

From this perspective, some commentators and researchers extrapolate that Bitcoin’s price should track electricity prices, natural-gas markets, or energy-sector equities, reasoning that miners’ marginal cost of production and geographic shifts in energy supply will be capitalized into BTC valuations. Others argue that the opportunity cost of diverting energy to mining, along with subsidies and externalities, should impose long-run constraints on Bitcoin demand and price levels. [ScienceDirect+2Earthjustice+2](#)

Despite the intuitive appeal of this narrative, empirical work on direct linkages between Bitcoin’s price and energy commodities remains limited and inconclusive. Studies of “clean cryptocurrencies” and portfolio diversification stress instead that environmental considerations mainly affect substitution within the crypto universe, not necessarily Bitcoin’s co-movement with oil or energy equities. [Nature+1](#)

3. Data and Methodology

3.1 Asset and Macro Data

The empirical analysis relies on a dataset comprising daily closing prices from early January 2012 through 17 November 2025 for the following tickers:

- **BTC** – Bitcoin spot price (USD)
- **Gold** – gold proxy (e.g., GLD ETF)
- **USTreasury** – U.S. 10-year Treasury total-return index or Treasury ETF
- **IXC** – global energy sector ETF
- **IYE** – U.S. energy sector ETF
- **VDE** – Vanguard energy ETF
- **SPY** – S&P 500 ETF (U.S. broad equities)
- **VXUS** – Vanguard Total International Stock ETF (ex-U.S. equities)

Each series spans 4,760 daily observations over the 2012–2025 period, with synchronized trading days to avoid spurious correlations arising from mismatched calendars.

In addition, the study employs monthly macroeconomic data:

- U.S. Consumer Price Index (CPI)
- U.S. Producer Price Index (PPI)
- Effective federal funds rate (FedFunds)
- 10-year U.S. Treasury constant-maturity yield (GS10)

These series are standard benchmark indicators used to evaluate inflation dynamics and the stance of monetary policy. [ScienceDirect+1](#) Monthly changes in these variables are matched to monthly asset returns.

3.2 Return Construction and Resampling

For each asset, daily log returns are computed as:

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right), r_t = \ln(P_t - 1P_t)$$

where P_t denotes the closing price on day t . The study also computes **weekly** and **monthly** log returns by resampling to end-of-week (W) and month-end (ME) prices, respectively, and applying the same log-difference transformation.

This approach yields three parallel correlation structures:

1. **Daily correlations**, emphasizing high-frequency co-movement and short-term risk relationships.
2. **Weekly correlations**, smoothing very high-frequency noise and aligning more closely with typical tactical trading horizons.
3. **Monthly correlations**, better suited for analyzing interactions with slowly moving macro variables such as inflation and policy rates.

3.3 Regime Definitions

To capture structural shifts in the macro-financial environment and Bitcoin’s market microstructure, the sample is segmented into four non-overlapping regimes:

1. **Early Adoption (2012–2015)** – characterized by low institutional participation, early exchange development, and relatively small market capitalization.
2. **Pre-COVID Expansion (2016–2019)** – period of rising institutional awareness, the 2017–2018 bull-bear cycle, and generally accommodative but “normalizing” post-crisis monetary policy.
3. **“QE-Everything” and Pandemic Period (2020–2021)** – massive fiscal and monetary stimulus, zero or negative real rates, and a surge in retail and institutional speculative activity.
4. **Tightening and Post-Pandemic Cycle (2022–2025)** – rapid interest-rate hikes to combat inflation, balance-sheet reductions, and subsequent stabilization, followed by renewed risk appetite under the Trump administration’s crypto-friendly policies and ETF-driven inflows. [Reuters+4ScienceDirect+4AP News+4](#)

For each sub-period, the study computes BTC’s correlation with each asset, using daily log returns restricted to the corresponding dates.

3.4 Rolling Correlations and Visualizations

To visualize time-varying relationships, the analysis calculates **rolling 252-day Pearson correlations** between BTC and each comparator asset, using one-year windows of daily returns. The resulting series are plotted in Figure 2, revealing how co-movement evolves across market cycles.

Figure 1 presents normalized price indices for BTC and the comparator assets, each rescaled to 1.0 at the beginning of the sample. This facilitates visual comparison of growth trajectories and volatility across assets over the 2012–2025 horizon.

3.5 BTC vs Macro Variables

To test the inflation-hedge and macro-hedge theses, the study computes correlations between **monthly BTC returns** and **monthly changes** in CPI, PPI, FedFunds, and GS10 yields. The macro data are differenced (for CPI and PPI) or expressed as changes in levels (for interest rates), and aligned with the asset return series.

This reduced-form approach complements the more sophisticated VAR and dynamic-correlation methods employed in the academic literature but is sufficient to test whether Bitcoin exhibits strong

contemporaneous relationships with core macro indicators that would justify its popular positioning as an inflation hedge or policy hedge. [Munich Personal RePEc Archive+3ScienceDirect+3Yonsei Pure+3](#)

4. Empirical Results

4.1 Full-Sample Correlations

The full-sample correlation matrix of daily log returns from 2012–2025 reveals several key features.

First, **Bitcoin's correlation with gold** is positive but modest, around 0.07 on a daily basis. BTC's correlation with U.S. Treasuries is essentially zero (around 0.00), while gold and Treasuries exhibit a much stronger positive relationship of about 0.31—consistent with gold's traditional role as a hedge against bond market stress and inflation uncertainty.

Second, **Bitcoin's correlations with equity benchmarks and energy-sector ETFs are higher:**

- BTC–SPY: ≈ 0.20 (daily)
- BTC–VXUS: ≈ 0.19
- BTC–IXC, BTC–IYE, BTC–VDE: ≈ 0.12 – 0.13 on a daily basis

These values are not extremely large but are substantially stronger than BTC's links to gold or Treasuries. On a monthly basis, the co-movement is even clearer: BTC–SPY correlation rises to roughly 0.35, BTC–VXUS to ≈ 0.26 , and BTC–energy sector correlations approach 0.19–0.20. Meanwhile, BTC–gold and BTC–Treasury correlations remain weak (≈ 0.07 and ≈ 0.02 , respectively).

Third, among the non-crypto assets themselves, **energy-sector ETFs exhibit near-unity correlations** (e.g., IXC–IYE–VDE around 0.98–0.99), as expected given their shared sector exposure. These very high intra-sector correlations set a useful benchmark: BTC's relationship with energy equities is materially lower than the correlation structure within the energy complex itself, undermining the claim that Bitcoin is simply an energy-linked proxy.

In sum, over the entire 2012–2025 sample, Bitcoin does **not** display the negative correlation with Treasuries that a classical hedge hypothesis would predict, nor does it co-move with gold in a manner consistent with reliable store-of-value status. Instead, it behaves as a moderately correlated, high-volatility extension of global equity risk.

4.2 Regime-Specific Correlations

Regime-specific correlations provide further insight into how Bitcoin's behavior has evolved.

4.2.1 Early Adoption (2012–2015)

During the early adoption phase, BTC correlations with all major asset classes are essentially indistinguishable from zero. BTC–gold, BTC–SPY, and BTC–VXUS correlations hover around ± 0.02 or less, and relationships with U.S. Treasuries and energy sector ETFs are similarly negligible.

This regime likely reflects a combination of low market depth, fragmented price discovery across exchanges, and idiosyncratic factors dominating BTC price movements. In such an environment,

Bitcoin was too small and too isolated from mainstream financial markets to exhibit stable cross-asset relationships.

4.2.2 Pre-COVID Expansion (2016–2019)

In the pre-COVID expansion, BTC correlations with other assets turn slightly positive but remain weak. BTC–SPY correlation is around 0.03; BTC–VXUS is near 0.02; BTC–gold is about 0.05. None of these magnitudes would justify describing Bitcoin as a dependable hedge or as tightly linked to global equities.

However, the shift from near-zero to modest positive co-movement suggests that **institutional awareness and increased integration into speculative portfolios** were beginning to align Bitcoin with broader risk sentiment.

4.2.3 “QE-Everything” and Pandemic Period (2020–2021)

The pandemic period marks a clear structural break. As central banks slashed policy rates, launched massive asset-purchase programs, and fiscal authorities delivered unprecedented stimulus, global markets experienced one of the most powerful liquidity booms in modern history. [ScienceDirect+1](#)

In this environment, **Bitcoin correlations with risk assets rise sharply:**

- BTC–SPY: ≈ 0.34
- BTC–VXUS: ≈ 0.36
- BTC–IXC: ≈ 0.27
- BTC–IYE: ≈ 0.23
- BTC–VDE: ≈ 0.23

BTC–gold correlation also increases to ≈ 0.17 , while BTC–Treasury correlation remains slightly negative (≈ -0.04). The magnitude and sign pattern are telling: BTC is **most closely aligned with equities and global energy risk**, not with gold or Treasuries.

The visual evidence in Figure 2 corroborates this story. Rolling 252-day correlations between BTC and SPY, VXUS, and the energy ETFs rise toward 0.4–0.5 during 2020–2021, while BTC’s relationships with gold and Treasuries remain weaker and more volatile. The pattern is consistent with Bitcoin acting as a **high-beta extension of equity risk** during a period when speculative appetite, especially from retail traders and leveraged institutions, was extraordinarily strong.

4.2.4 Tightening and Post-Pandemic Cycle (2022–2025)

The post-pandemic tightening cycle initially saw aggressive rate hikes and quantitative tightening as central banks attempted to rein in inflation. However, by 2024–2025 the policy stance had partially normalized, while the Trump administration adopted explicitly crypto-friendly policies and established the Strategic Bitcoin Reserve. [The Washington Post+4ScienceDirect+4White House+4](#) Spot Bitcoin ETFs launched and quickly attracted record inflows, contributing to dramatic price appreciation. [Investopedia+4The Economic Times+4AP News+4](#)

In this regime, BTC’s correlations with risk assets remain elevated or increase further:

- BTC–SPY: ≈ 0.39
- BTC–VXUS: ≈ 0.36
- BTC–IXC: ≈ 0.18 – 0.19
- BTC–IYE: ≈ 0.19
- BTC–VDE: ≈ 0.19

BTC–gold correlation is about 0.11, while BTC–Treasuries correlation is slightly positive at ≈ 0.02 . The main message is that **Bitcoin remains firmly in the risk-asset camp**, with correlations to equities far stronger than to classical hedges.

Notably, the recent 2025 correction from all-time highs above \$120,000 coincides with emerging stress in private-credit markets and concerns about liquidity conditions, as illustrated by the First Brands bankruptcy and renewed scrutiny of leveraged finance. [Managed Funds Association+7Private Debt Investor+7Jefferies.com+7](#) The correction does not align with any major discrete shock to gold or energy markets, again suggesting that **changes in financial conditions and risk appetite, rather than commodity prices or Treasury dynamics, are the primary drivers of BTC’s price path.**

4.3 Rolling 252-Day Correlations

Figure 2 illustrates rolling one-year correlations between Bitcoin and each comparator asset.

Two patterns stand out:

1. **Pre-2020, correlations fluctuate around zero**, with occasional short-lived spikes but no persistent relationship with any single asset. This is consistent with Bitcoin’s small market size and limited integration into institutional portfolios.
2. **From 2020 onward, correlations with SPY, VXUS, and energy ETFs trend upward**, reaching peaks near 0.5–0.6 at various times, especially during the late-pandemic boom and ETF-driven rallies. By contrast, BTC–gold and BTC–Treasury correlations remain comparatively low and more unstable, oscillating around small positive or slightly negative values.

The rolling-correlation plot therefore reveals a **structural shift from isolation to integration**: Bitcoin has evolved from an idiosyncratic niche asset into a high-beta component of the broader risk complex, particularly during periods of abundant liquidity and speculative fervor.

4.4 BTC vs Macro Variables

To assess Bitcoin’s relationship with macroeconomic fundamentals, the study examines correlations between **monthly BTC returns** and **monthly changes in CPI, PPI, FedFunds, and GS10**.

The results are strikingly modest:

- BTC–CPI: ≈ -0.02
- BTC–PPI: ≈ 0.04

- BTC–FedFunds: ≈ -0.11
- BTC–GS10: ≈ 0.08

In other words, **Bitcoin's monthly price changes are only weakly related to contemporaneous inflation and interest-rate dynamics**. Small negative correlation with FedFunds is consistent with the notion that easier monetary policy tends to support risk assets, but the magnitude is far from what would be expected if BTC were a direct macro hedge.

These findings dovetail with the academic literature, which suggests that Bitcoin may exhibit inflation-hedging characteristics in certain regimes or horizons but does not behave like a traditional inflation hedge such as TIPS or commodities, and certainly not with the stability associated with gold. [Munich Personal RePEc Archive+3ScienceDirect+3Yonsei Pure+3](#)

5. Evaluating the Popular Investment Theses

5.1 Bitcoin as a Hedge Against U.S. Treasuries

A hedge against Treasuries would, in principle, display **consistently negative correlation with Treasury returns** and possibly positive correlation with inflation and policy-rate expectations. The evidence presented above does not support this claim:

- Full-sample BTC–Treasury correlation is essentially zero on a daily basis and close to zero on a monthly basis.
- Regime-specific correlations are small in magnitude, occasionally slightly negative (e.g., -0.04 during 2020–2021) but more often near zero or slightly positive.
- BTC–Treasury correlations do not exhibit the strong and stable pattern observed between gold and Treasuries.

Furthermore, during the sharp risk-off episodes of March 2020 and various post-2021 equity drawdowns, Bitcoin often declined alongside equities and other risk assets rather than offsetting Treasury losses or delivering diversifying gains. This behavior aligns with other research finding that Bitcoin’s safe-haven characteristics, when present, are short-lived and largely confined to specific tail states. [ScienceDirect+3Taylor & Francis Online+3ResearchGate+3](#)

The hedge thesis may have gained traction due to **selective observation**: early episodes when Bitcoin rallied during political turmoil or idiosyncratic currency crises were generalized into a structural claim, even though the broader time series does not support such a conclusion. Media narratives that highlight eye-catching price spikes during crises, while ignoring periods when BTC sells off alongside risky assets, have reinforced this misperception.

In light of the evidence, Bitcoin is better described as a **complementary risk asset** in portfolios that already hold equities and high-yield credit, rather than as a Treasury hedge. Portfolio-level benefits, if any, derive from BTC’s asymmetric payoff and idiosyncratic drivers, not from systematically offsetting bond-market risk.

5.2 Bitcoin as Digital Gold and Store of Value

The digital-gold thesis rests on the idea that Bitcoin, like physical gold, is a scarce monetary asset with limited supply, robust to debasement by any single government. This argument has been elaborated by institutional reports that position Bitcoin as an “aspirational store of value” and potential long-term portfolio anchor. [Lopp+5Fidelity Digital Assets+5Fidelity Digital Assets+5](#)

However, to function as a **store of value akin to gold**, Bitcoin would need to exhibit at least two properties in realized market data:

1. **Low or negative correlation with risk assets**, particularly equities, so that it preserves wealth when other assets experience drawdowns.
2. **Reasonable stability of purchasing power** across macro regimes, avoiding extreme drawdowns that wipe out large fractions of wealth for long periods.

Neither condition is currently satisfied.

- Bitcoin's correlations with SPY and VXUS are consistently higher than its correlations with gold or Treasuries, especially since 2020. BTC moves with risk assets more than with traditional stores of value.
- The normalized price chart in Figure 1 underscores Bitcoin's extreme volatility: multiple drawdowns of 70–80% or more have occurred within the sample, followed by parabolic rallies. Such a profile may be attractive for speculative investment but is difficult to reconcile with a conservative store-of-value role.

Academic work on Bitcoin's inflation-hedging properties further complicates the picture. While several studies document that Bitcoin tends to appreciate in response to inflation shocks, they emphasize strong context dependence and unstable relationships across countries and time periods.

[ORBilu+5ScienceDirect+5Yonsei Pure+5](#) Some even argue that Bitcoin's inflation-hedging properties have weakened as it has become more financialized and subject to liquidity-driven cycles.

The **digital-gold narrative nevertheless persists**, in part because it serves multiple constituencies:

- For early adopters and ideological supporters, it provides a powerful rhetorical frame linking Bitcoin to centuries-old monetary history.
- For institutional allocators, it offers a familiar conceptual box—"alternative store of value"—that makes BTC easier to slot into existing portfolio frameworks.
- For policymakers advocating strategic reserves, it provides justification for treating Bitcoin as analogous to gold in a national balance sheet context. [Reuters+3Wikipedia+3The Washington Post+3](#)

Yet until Bitcoin exhibits a more stable volatility profile and less pronounced co-movement with equities, describing it as a store of value **in practice** remains aspirational rather than descriptive. At present, Bitcoin behaves more like a highly volatile, liquidity-sensitive risk asset than like a conservative monetary reserve.

5.3 Bitcoin as a Reflection of Energy Markets

The energy-linked thesis posits that Bitcoin's price is tied to electricity demand and energy-sector dynamics, because proof-of-work mining consumes substantial amounts of power. Estimates from the Cambridge Bitcoin Electricity Consumption Index and related studies suggest that Bitcoin mining accounts for a non-trivial share of global and U.S. electricity consumption. [Nature+9CCAF+9U.S. Energy Information Administration+9](#)

From these premises, some commentators conclude that Bitcoin is effectively a derivative on electricity or natural gas prices, or that energy-sector equities should exhibit strong co-movement with BTC. The empirical evidence here is mixed but, on balance, does not support a strong equivalence:

- Over the full sample, BTC–energy sector correlations are **modest** (≈ 0.12 daily, ≈ 0.19 monthly) and clearly lower than the correlations among the energy ETFs themselves (≈ 0.98 – 0.99).

- In regime-specific analyses, BTC–energy correlations do increase during periods of booming risk appetite (2020–2021, 2023–2025), but so do BTC–SPY and BTC–VXUS correlations. There is little evidence that energy variables are uniquely important relative to general equity risk.
- Rolling correlations do not show a persistent structural tie between BTC and energy equities; instead, co-movement rises and falls alongside broader market cycles.

These patterns suggest that **energy demand is a necessary but not sufficient condition** for Bitcoin’s valuation. Mining costs and electricity prices may set long-run viability thresholds for miners and may influence geographic shifts in hash power, but the marginal price of BTC in liquid markets appears much more closely tied to **financial liquidity, risk appetite, and speculative demand** than to the direct cost of electricity alone.

Environmental and regulatory discourse around crypto mining also tends to focus on externalities, subsidies, and grid stress rather than on any claim that Bitcoin’s price mirrors energy prices.

[Nature+5U.S. Energy Information Administration+5Utility Dive+5](#) In other words, the energy-linked thesis has more traction as a **policy critique** than as a rigorous pricing model.

6. Bitcoin as a Global Liquidity and Risk Barometer

If Bitcoin is not reliably a Treasury hedge, digital gold, or energy derivative, what does the evidence suggest instead?

The correlation patterns, regime shifts, and macro relationships point toward an alternative characterization:

Bitcoin operates as a high-beta barometer of global liquidity, speculative risk appetite, and the health of leveraged finance.

Several strands of evidence support this interpretation.

6.1 Sensitivity to Liquidity Regimes

During the 2020–2021 pandemic period, unprecedented monetary and fiscal stimulus flooded financial markets with liquidity, pushed real yields deeply negative, and spurred a broad speculative boom in equities, options, SPACs, and other high-beta assets. In this environment, Bitcoin’s correlations with SPY and VXUS surged, and BTC delivered some of its most explosive gains.

Conversely, when central banks tightened policy aggressively in 2022–2023, Bitcoin experienced sharp drawdowns alongside growth equities and other rate-sensitive risk assets. The relationship between BTC returns and changes in the federal funds rate, while modest in magnitude, is directionally consistent with this narrative: Bitcoin tends to struggle when policy is tightening and liquidity is being withdrawn. [ScienceDirect+1](#)

6.2 ETF Inflows, Strategic Reserves, and Financialization

The launch and rapid growth of spot Bitcoin ETFs in 2024–2025 dramatically increased Bitcoin’s accessibility to traditional investors. Record inflows into these vehicles have been closely associated with surges in BTC’s price to successive all-time highs above \$111,000, \$118,000, and later around \$125,000. [institutional.fidelity.com+6The Economic Times+6AP News+6](#)

At the same time, the U.S. government’s establishment of a Strategic Bitcoin Reserve and similar discussions at other central banks have lent Bitcoin symbolic status as a quasi-reserve asset and “digital Fort Knox,” even though these initiatives are largely based on seized assets rather than active purchases. [The Guardian+7White House+7Wikipedia+7](#)

Together, these developments represent **deepening financialization**: Bitcoin is increasingly intermediated through regulated vehicles and integrated into multi-asset portfolios. Such integration naturally increases co-movement with other financial assets as common factors—liquidity, leverage, and risk sentiment—dominate idiosyncratic drivers.

6.3 Interaction with Private Credit and Leveraged Finance

The recent Bitcoin rally and subsequent correction have unfolded against a backdrop of rising concerns in private-credit and leveraged-finance markets. The bankruptcy of First Brands Group, the associated exposure of funds at UBS and Jefferies, and broader debates about the resilience of private credit illustrate how liquidity stress can emerge in ostensibly “niche” corners of the market and propagate

through balance sheets. [Nature+15Private Debt Investor+15Jefferies.com+15](#) Energy variables thus enter Bitcoin's valuation indirectly—by affecting regulatory risks, production costs, and public perception—rather than through a persistent correlation structure akin to that between oil and energy equities.

7. Why the Flawed Theses Became Popular

If the data do not support Bitcoin as a Treasury hedge, digital gold, or energy derivative, why have these narratives become so entrenched?

Several mechanisms can be identified.

7.1 Narrative Simplicity and Marketing

Each thesis offers a **simple, intuitive frame** that can be easily communicated:

- “Digital gold” analogizes Bitcoin to a familiar asset with centuries of monetary history.
- “Hedge against Treasuries” aligns with distrust of fiscal policy and concerns about debt sustainability.
- “Backed by energy” resonates with the physical-scarcity intuition and the visible electricity footprint of mining.

These frames are powerful marketing tools for exchanges, asset managers, and advocacy groups seeking to attract attention and capital. Institutional whitepapers, fund prospectuses, and influencer commentary routinely deploy such labels, sometimes supported by selective or short-horizon empirical evidence. [YouTube+5Fidelity Digital Assets+5Fidelity Digital Assets+5](#)

7.2 Regime-Selective Evidence

Many early studies and commentaries used **limited samples**—for example, focusing on post-2016 data or narrow crisis windows such as 2020Q1. In such periods, it is possible to find temporary episodes where Bitcoin appears to behave like a hedge or safe haven, especially when comparisons are drawn against specific currencies or indices. [ResearchGate+3ORBilu+3ScienceDirect+3](#)

Over time, as the dataset lengthens and additional regimes (e.g., post-pandemic tightening, ETF financialization) are observed, the initial conclusions must be revisited. However, narratives and marketing copy tend to exhibit **path dependence**: once an asset is branded as “digital gold,” it is costly for institutions to abandon that frame, even when newer evidence contradicts it.

7.3 Ideological and Political Alignment

Bitcoin sits at the intersection of technology, finance, and politics. For some constituencies, it symbolizes resistance to central banks, inflationary policies, or perceived financial repression. For others, it represents technological innovation and economic opportunity.

The adoption of Bitcoin-friendly policies by the Trump administration—culminating in the Strategic Bitcoin Reserve and broader deregulatory steps—has further politicized the asset, with supporters and critics often selecting narratives that align with their ideological priors. [The Washington Post+5Wikipedia+5White House+5](#) In such an environment, **empirical nuance can be overshadowed by rhetorical utility**.

7.4 Media Incentives and Volatility

Bitcoin's dramatic price swings and frequent all-time highs generate significant media attention.

Articles highlighting short-term co-movement with gold, Treasuries, or energy prices during specific weeks or months are more likely to attract readers than nuanced discussions of long-run correlations.

Moreover, because Bitcoin is so volatile, **cherry-picking time windows** can easily produce apparent relationships that vanish when inspected over longer horizons. This “narrative volatility” mirrors price volatility and contributes to the persistence of flawed theses.

8. Implications and Conclusion

The empirical analysis presented in this paper, using daily, weekly, and monthly data from 2012–2025 across multiple market regimes, leads to several clear conclusions:

1. **Bitcoin is not a reliable hedge against U.S. Treasuries or a robust safe haven.** Correlations with Treasury returns are near zero, and during major risk-off episodes Bitcoin has often sold off alongside equities.
2. **Bitcoin does not yet function as a realized store of value akin to gold.** While narratives and some studies emphasize inflation-hedging potential, realized correlations with inflation and macro variables are weak, and the asset's extreme volatility and large drawdowns are at odds with traditional store-of-value behavior.
3. **Bitcoin is not tightly bound to energy markets in the manner of a commodity derivative.** Although mining consumes meaningful amounts of electricity and raises legitimate policy concerns, BTC's price co-moves more strongly with broad equity indices than with energy-sector equities, and there is little evidence of direct linkage to oil or electricity prices.
4. **Bitcoin behaves primarily as a liquidity-sensitive, high-beta risk asset.** Its correlations with global equities have increased significantly since 2020, particularly during periods of abundant liquidity, ETF inflows, and speculative enthusiasm. Rolling correlations and regime analysis show BTC acting as an amplified expression of broader financial conditions, rather than as a diversifying hedge.
5. **Recent market events are consistent with this barometer interpretation.** The 2024–2025 rally to new all-time highs, fueled by record ETF inflows and supportive policy signals, followed by sharp corrections amid emerging strains in private credit and leveraged finance, illustrate how Bitcoin responds more to changes in liquidity and risk appetite than to inflation prints or energy prices. [Apollo Academy+9Reuters+9etf.com+9](#)

These findings do not imply that Bitcoin is devoid of value or strategic relevance. Rather, they suggest that **Bitcoin's primary role in portfolios and policy debates should be reframed**. Instead of treating BTC as a direct hedge against sovereign debt, inflation, or energy prices, analysts and investors should recognize it as a **global risk-sentiment thermometer**—a liquid, continuously traded instrument that reflects and amplifies changes in speculative appetite and financial conditions.

For portfolio construction, this implies that:

- Allocations to Bitcoin should be sized according to risk tolerance and viewed as part of the **equity and high-beta risk budget**, not as a substitute for Treasuries or gold.
- Risk management frameworks should recognize BTC's tendency to **sell off during liquidity shocks**, rather than expecting it to provide ballast when traditional assets are stressed.
- Any diversification benefits likely stem from **non-linear payoff structures and idiosyncratic catalysts**, not from systematic negative correlation with other assets.

For policymakers, the barometer view underscores that:

- Bitcoin's price can serve as a **real-time indicator of speculative excess or stress** in the financial system, complementing more traditional measures such as credit spreads and volatility indices.
- Concerns about energy consumption and environmental impact remain valid and should be addressed through targeted regulation and pricing of externalities rather than through assumptions about BTC's macro-hedging role. [Nature+6CCAF+6U.S. Energy Information Administration+6](#)

Finally, for researchers, the results highlight the importance of **long-horizon, regime-aware empirical work**. As Bitcoin and other digital assets become increasingly integrated into mainstream finance, their behavior is likely to continue evolving. The early adoption phase, the pandemic liquidity boom, and the ETF-driven era each exhibit distinct correlation structures. Future work should therefore employ methods that can capture such non-stationarity—dynamic factor models, state-space representations, and regime-switching frameworks—rather than relying on static full-sample correlations.

In conclusion, the evidence assembled here supports a re-interpretation of Bitcoin's place in the financial ecosystem. Rather than digital gold, anti-Treasury hedge, or energy proxy, Bitcoin is best understood, at this stage of its evolution, as a **global liquidity and speculative-risk barometer**—a role that is both powerful and precarious, and one that should be recognized clearly in both investment practice and public discourse.

Disclaimer: This paper is for informational and educational purposes only and does not constitute investment, legal, or tax advice. Digital assets are highly volatile and may not be suitable for all investors.

Appendix

Source code to generate the analysis may be found in:

https://github.com/ryan-aday/Bitcoin_Correlation_Study

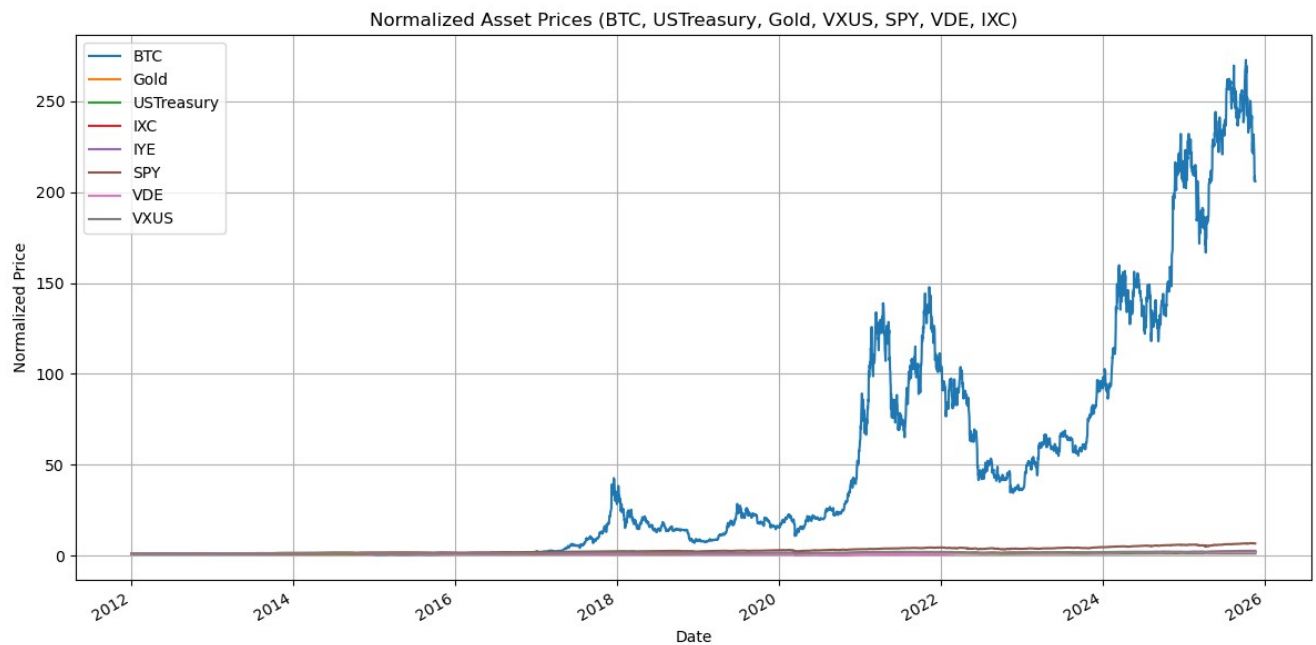


Figure 1. Normalized Asset Prices from 2012 to 2025

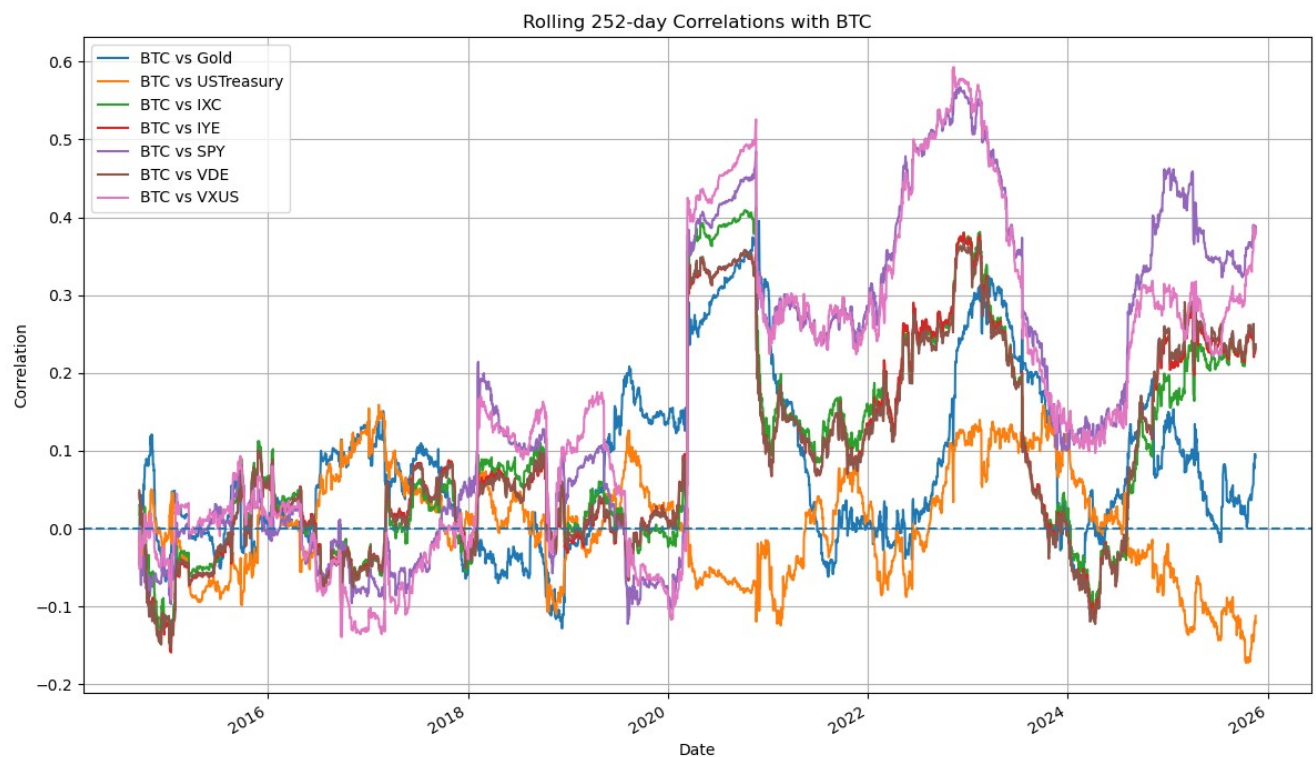


Figure 2. Rolling 252-Day Correlations with BTC

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